

ARTICLE

Supplementary Material for “Energy-Gated DA-TPP: Batch Active Learning for Early Target Recovery in Generated Crystal Pools”

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ABSTRACT: This Supplementary Material reports the parameters, source tables, and robustness analyses supporting the main paper. Energy-Gated DA-TPP recovered seven more surrogate targets than Greedy within the first 80 queries and retained a 14.7% normalized-AUTC advantage through 160 queries. An acquisition-blind post-selection audit of 20 historical DFT attempts yielded a nested leave-one-out ROC–AUC of 0.820 and assigned the early Gate prefix a larger expected evaluable-target count. CHGNet and MACE-MP completed all 640 candidates, and four generated LiCr₂O₄ candidates completed the stated PAW–PBE+*U* relaxation–static workflow. Exact run-identity audits, source values, arithmetic checks, crystallographic descriptors, and a ten-seed Materials Project Mn-oxide control are retained so that model estimates remain distinguishable from observed labels and DFT records. Development-cohort threshold and weight sweeps, together with same-protocol Greedy, MC-dropout, and random controls, provide the parameter and baseline context for the main discussion. A development-selected $\gamma = 0.05$ protocol was then fixed before a ten-seed held-out Gate–Greedy comparison, in which mean AUTC through 160 queries increased from 0.2967 to 0.3223.

KEYWORDS: Supplementary Material; active learning; DFT evaluability; acquisition-blind evaluation; reproducibility

S1 Supplementary evidence overview

The supplementary evidence supports a focused early-budget interpretation. In the protocol-matched archive, Gate recovered 26 surrogate targets within 80 queries compared with 19 for Greedy. Its normalized AUTC was 12.2% higher at an 80-query stopping horizon and 14.7% higher through 160 queries. Both methods ultimately recovered the same 78-target set, so the observed advantage concerns the earlier placement of useful candidates rather than a higher endpoint ceiling.

The downstream records provide two additional assessments of that ordering result. The acquisition-blind evaluator assigned the 80-query Gate prefix 16.63 expected DFT-evaluable targets, compared with 10.99 for Greedy, without influencing selection. Separately, both MLIPs completed the full 640-candidate pool and all four reported LiCr₂O₄ candidates completed the stated first-principles workflow.

The following sections provide the exact inputs, tables, provenance, and scope limits for these results, including the independently initialized held-out comparison.

S2 Target task and evidence types

The analysis is defined by the 640-candidate Li–M–O pool, its identifiers, the ALIGNN reference table, and the stored acquisition histories. The interval $[-2.18, -2.02]$ eV atom⁻¹ contains 78 surrogate targets. It belongs to the archived query task rather than to a stability or application criterion, and it is fixed independently of the DFT-evaluability metric.

Retained structures and candidate identifiers establish the MatterGen-derived provenance of the pool. Recoverable request fields are retained for provenance, whereas acquisition analysis uses only the frozen candidate identities, ALIGNN reference table, and stored query histories. No cross-convention conversion is applied among generation, ALIGNN, and current DFT energies.

The source files separate three evidence types:

1. observed CGCNN query histories and archived DFT records;
2. an acquisition-blind machine-learning endpoint for strict DFT evaluability;
3. physical properties outside the present target-recovery and workflow-feasibility scope.

Table S1: Notation and evidence layers used in the frozen-pool analysis.

Symbol	Definition	Evidence status
C, N	Frozen candidate pool and its size ($N = 640$)	archived objects
y_i, z_i	ALIGNN reference value and interval-membership label	reference-table oracle
p_i, e_i	CGCNN interval-hit probability and DFT-evaluability score	ML estimates
d_i	Strict completion label for a historical DFT attempt	archived DFT record
B_t, b, Q	Round- t batch, batch size 16, and query budget	protocol
$Y_a(q), H_a(q)$	Recovered targets and DFT-evaluability endpoint sum	history; ML endpoint
G_t, M_t	Group concentration and standardized front margin	gate diagnostics

S3 Archived implementation and run identifiers

The recovered parameter file and claim map are presented first so that every later numerical statement can be traced to a frozen configuration and evidence type.

Table S2: Recoverable settings of the archived Energy-Gated DA-TPP runs.

Setting	Value	Setting	Value
M_0	1.00	G_0	0.50
α	0.10	β	0.20
γ	0.10	Acquisition batch	16
MC passes	3	Dropout rate	0.30
Refit epochs	10	Learning rate	0.001
CGCNN batch size	256	Total query budget	640

Note: Values are read from the archived run configurations and remain fixed within the stored histories. Development-cohort sensitivity and the separately held-out protocol are reported in Tables S12–S14.

Implementation seeds 5–14 identify the formal archive. Main-text run IDs 1–10 map to them in ascending order. The original values remain in `SourceData/paired_autc_at_320.csv` and the history files.

Table S3: Evidence layers and their roles in the study.

Evidence	Quantity assessed	Role in the study
Frozen query histories	Recovery of ALIGNN reference targets	Primary surrogate-ordering evidence
Frozen seeds 15–24 paired histories	Early AUTC and checkpoint recovery under $\gamma = 0.05$	Held-out robustness assessment
Acquisition-blind e_i scores	Estimated DFT evaluability at fixed checkpoints	Post-selection downstream assessment
Four completed Cr workflows	Completion, forces, geometry, moments, and energies	First-principles feasibility examples
Archived-prefix hash audit	Distinct archived query prefixes	Run-identity and provenance check
Selector-equivalence audit	Recorded route changes under the two diagnostics	Mechanism and fallback analysis

The settings in Table S2 were loaded before each archived run and did not change. They are reported as recovered implementation values; the separate development-cohort sensitivity analysis is given in Table S12.

S4 Acquisition-blind DFT-evaluability assessment

Strict label one requires electronic and ionic convergence, a separate static calculation, and a reproducible formation energy. Ten of the 20 archived attempts meet all four requirements. Seven other records lack a reconstructable energy, two contain a VASP or electronic failure, and one has an unconverged static step.

Table S4: Nested leave-one-out assessment of the DFT-evaluability classifiers.

Model	ROC–AUC	Balanced accuracy	Brier score	Log loss
Shallow gradient boosting	0.820	0.700	0.183	0.545
Shallow random forest	0.660	0.750	0.234	0.664
Regularized logistic regression	0.540	0.600	0.258	0.713
Laplace logistic model	0.460	0.600	0.282	0.766

Note: The development archive contains 20 DFT attempts, including 10 strictly evaluable records. Preprocessing and model selection are performed inside each outer training fold.

Shallow gradient boosting achieved the highest nested leave-one-out ROC–AUC, 0.820. Across the 20-record archive, this corresponds to correct ordering in 82% of positive–negative record pairs used by the ROC calculation. The model is used only as an acquisition-blind post-selection audit.

Candidate-level labels are stored in `SourceData/historical_dft_binary_labels.csv`; the complete cross-validation table is in `SourceData/dft_evaluability_model_cv.csv`. The selected shallow gradient-boosting model uses a fixed 0.50 probability threshold. Candidate IDs and historical policy ranks are excluded from the features.

S5 Early-prefix DFT-evaluability audit

The following table applies the frozen evaluator to the already selected Gate and Greedy prefixes at the four reported stopping horizons.

At 80 queries, Gate had recovered 26 targets with a summed expected evaluable count of 16.63, whereas Greedy had recovered 19 targets with an expected count of 10.99. The same early prefix also gave the larger thresholded count, 20 versus 12. The difference narrows as Greedy recovers the deferred targets and disappears once both policies reach the common endpoint.

For a target candidate i , the expected contribution is e_i , while the thresholded contribution is $\mathbb{I}(e_i \geq 0.50)$. A non-target contributes zero to both target-yield endpoints. Complete overlays and score provenance are retained in `SourceData/paired_dft_evaluability_detail.csv` and `SourceData/dft_evaluability_scores.csv`.

Table S5: Surrogate target recovery and the formal DFT-evaluability endpoint at the fixed query checkpoints.

Queries	Method	Target hits	Expected evaluable	$e_i \geq 0.50$	Mean e_i
80	Gate	26	16.63	20	0.64
	Greedy	19	10.99	12	0.58
160	Gate	54	33.87	41	0.63
	Greedy	54	31.60	35	0.59
240	Gate	67	39.93	46	0.60
	Greedy	69	40.96	47	0.59
320	Gate	78	46.51	53	0.60
	Greedy	78	46.51	53	0.60

Note: Expected evaluable is $\sum z_i e_i$ over the targets already selected by the CGCNN policy. The acquisition-blind e_i values are ML estimates and are not supplied to Gate or Greedy.

S6 Query-prefix and target-set audit

70

Table S6: Identity audit of the ten archived records within each method.

Method	Queries	Stored records	Unique complete prefixes	Unique target sets
Gate	80	10	1	1
Gate	160	10	1	1
Gate	240	10	1	1
Gate	320	10	10	1
Greedy	80	10	1	1
Greedy	160	10	1	1
Greedy	240	10	1	1
Greedy	320	10	10	1

Note: At 320 queries, complete prefixes differ only in non-target ordering. All records within a method recover the same target set. Prefixes through 240 queries are identical within each method.

The hash-level audit is retained in `SourceData/target_set_identity_audit.csv` and `SourceData/target_set_identity_summary.csv`. Because each method has one distinct complete prefix through 240 queries, the stored IDs do not provide replicates for an inferential paired test.

71

72

73

S7 Reconstructed early-horizon AUTC advantage

74

Table S7: Normalized AUTC for the canonical recoverable Gate and Greedy trajectories.

Stopping budget	Gate AUTC	Greedy AUTC	Relative Gate difference
80	0.1410	0.1256	+12.2%
160	0.3205	0.2795	+14.7%
240	0.4692	0.4504	+4.2%
320	0.5827	0.5718	+1.9%

Note: AUTC uses the left-continuous 16-query-batch sum and the 78-target ceiling. The relative difference is $(\text{Gate}/\text{Greedy} - 1) \times 100\%$.

Gate exceeded Greedy in normalized AUTC at every reported stopping horizon. The relative differences were 12.2%, 14.7%, 4.2%, and 1.9% through 80, 160, 240, and 320 queries, respectively. The largest advantage therefore occurs in the early-budget regime targeted by the method.

75

76

77

Candidate-level histories were recomputed with a left-continuous 16-query-batch sum at stopping budgets of 80, 160, 240, and 320. Input trajectories and values are retained in `SourceData/paired_cgcnn_histories.csv` and `SourceData/v50_autc_checkpoints.csv`. The 20 stored 320-query row values and implementation seed mapping remain in `SourceData/paired_autc_at_320.csv`. Differences after the identical early prefixes do not create independent replicates.

S8 Complete full-pool MLIP execution

Both low-fidelity models were executed for every candidate. Figure S1 shows the resulting per-candidate wall-time distributions in the recorded environment.

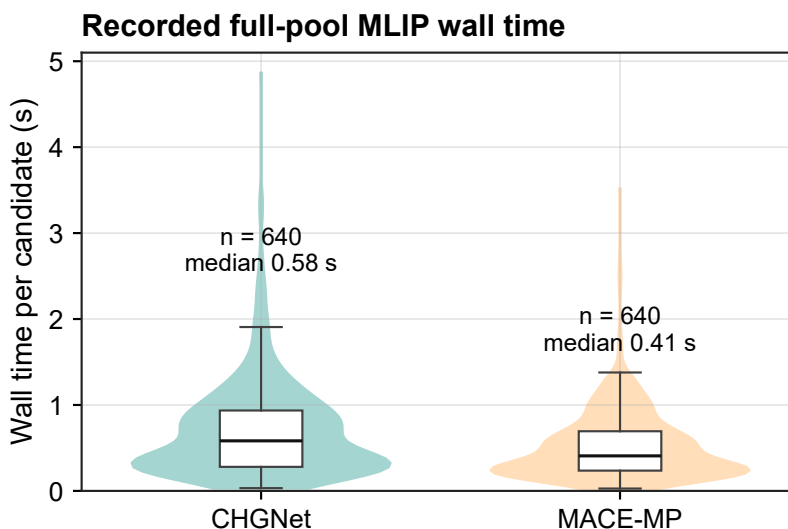


Figure S1: Per-candidate wall times for 640 CHGNet and 640 MACE-MP evaluations. Boxes show the interquartile range and median within each violin. Values come from the archived MLIP environment and are not normalized to DFT hardware.

The two models produced 1,280 complete records in 781.545 s of aggregate wall time. Median per-candidate values were 0.582 s for CHGNet and 0.408 s for MACE-MP. Candidate-level values are supplied in `SourceData/mlip_full_pool_results.csv`, whose SHA-256 is 37fc2a9ce0f7eb9c4d914b5f347f0c127a32754edb1bc38c06748f7fb7a51892.

For the ten records with reconstructable DFT energies, leave-one-out offset correction of MACE-MP produced an MAE of 0.0145 eV atom⁻¹ and an RMSE of 0.0159 eV atom⁻¹. Seven observed DFT values fell inside the nominal 95% intervals. Candidate-level outer-fold predictions are retained in `SourceData/energy_calibration_best_oof.csv`.

S9 Descriptive six-policy archive

The legacy archive preserves one complete trajectory for each of the six implemented policies. Its checkpoint counts and evaluator overlay are reported below as descriptive evidence.

Table S8: Descriptive DFT-evaluability endpoint on the single archived six-policy histories.

Method	80 queries		160 queries		240 queries		320 queries	
	Target	Exp. eval.	Target	Exp. eval.	Target	Exp. eval.	Target	Exp. eval.
Energy-Gated DA-TPP	33	20.20	50	29.78	69	40.66	78	46.51
Explore	29	17.32	44	26.05	60	35.44	74	43.60
Predicted-Target Greedy	27	16.36	47	28.78	69	40.96	78	46.51
MC Dropout	8	4.44	27	14.42	36	20.43	49	28.56
Modulus/Gradient-Norm	10	5.54	20	9.22	39	18.83	58	32.40
Random Sampling	6	4.29	18	11.30	26	15.57	39	22.73

Note: Each row is one historical trajectory and is provided only as descriptive context. Exp. eval. is the expected number of target candidates scored as DFT-evaluability by the acquisition-blind endpoint. These histories are not paired replicates of the formal Gate-Greedy archive.

Gate and Greedy in Table S8 are historical single-run records, distinct from the matched archive used for the primary comparison. They preserve the older comparison with Explore, MC Dropout, Modulus/Gradient-Norm, and Random Sampling. The table is descriptive.

S10 First-principles protocol for four completed candidates

The first-principles set contains C079-1, C126-0, C196-1, and C234-3. These four records are separate from the 20 attempts used by the DFT-evaluability model and were not sampled as a Gate-Greedy yield comparison.

All four calculations used VASP 6.5.1, current PAW-PBE datasets, Dudarev PBE+ U with $U_{\text{eff}}(\text{Cr}) = 3.7$ eV, and ENCUT= 649 eV. Relaxations used EDIFF= 10^{-6} eV, EDIFFG= -0.05 eV $\text{\AA}^{-1}$, IBRION= 2, ISIF= 3, NSW= 200, ISPIN= 2, ISMEAR= 0, SIGMA= 0.05 eV, LASPH=TRUE, and LMAXMIX= 4. Independent statics used the relaxed structure with IBRION= -1 and NSW= 0. Gamma-centred meshes were generated with an approximately 0.15 $\text{\AA}^{-1}$ reciprocal-spacing rule: $15 \times 11 \times 7$ for C079-1, $15 \times 10 \times 9$ for C126-0, and $16 \times 16 \times 5$ for C196-1 and C234-3.

POTCAR text is not redistributed. The archived SHA-256 identifiers are 201875120238865c...fbb95b for Li, 836672959fc86f3b...9bd744b for Cr, and 8a74b9a1f5fdb3d...a3eea64 for O; the complete values are retained in SourceData/DFT_completed_candidates_report_zh.md.

S11 Completed first-principles workflows and energy reconstruction

All four LiCr_2O_4 candidates completed full cell-and-ion relaxation and a separate static calculation. Their reconstructed formation energies span -2.1676 to -2.0336 eV atom^{-1} , maximum residual forces remain below the 0.05 eV $\text{\AA}^{-1}$ stopping criterion, and no automated cell-collapse flag was raised.

Table S9: Completed LiCr_2O_4 first-principles feasibility calculations.

Candidate	E_f (eV atom^{-1})	F_{max} (eV $\text{\AA}^{-1}$)	ΔV (%)	Shortest distance ($\text{\AA}$)	Moment (μ_B cell $^{-1}$)	Gap (eV)
C079-1	-2.164905884	0.045128	$+2.918$	1.917646	4.937	0.0035
C126-0	-2.167618531	0.034407	-1.358	1.938847	4.922	0.0010
C196-1	-2.033635033	0.033610	$+0.962$	1.724317	4.864	0.0004
C234-3	-2.033665270	0.044104	$+0.586$	1.724466	4.864	0.0015

Note: The reported gap is a near-zero-gap diagnostic. The magnetic moment belongs to the tested initialization and is not a magnetic-ground-state assignment.

The internally consistent formation energy is

$$E_f = \frac{E_{\text{LiCr}_2\text{O}_4} - \mu_{\text{Li}} - 2\mu_{\text{Cr}} - 2E_{\text{O}_2}}{7}. \quad (1)$$

The retained reference energies are $\mu_{\text{Li}} = -1.90574968$ eV atom⁻¹, $\mu_{\text{Cr}} = -6.16417684$ eV atom⁻¹, and $E_{\text{O}_2} = -9.89314292$ eV molecule⁻¹. Two independent recalculation scripts agree within 10⁻¹⁵ eV atom⁻¹. The check establishes arithmetic consistency inside this protocol. Because no validated compatibility mapping to Materials Project or ALIGNN energies is available, no cross-scale conversion is applied (archived status: same_scale_status=UNRESOLVED).

All four relaxations satisfy $F_{\text{max}} \leq 0.05$ eV Å⁻¹ without an automated cell-collapse flag, and all four final static calculations completed from converged relaxed structures. Restart provenance and the recorded C234-3 lattice warning are retained in the source archive. The calculations assess workflow completion and local relaxation under the stated protocol; phase stability requires a separate competing-phase analysis.

Final structures were parsed from the retained static `vasprun.xml` records and exported as `Structures/C079-1_final.cif`, `Structures/C126-0_final.cif`, `Structures/C196-1_final.cif`, and `Structures/C234-3_final.cif`. The source XML path, XML hash, CIF hash, lattice parameters, site count, and final volume are listed in `SourceData/v50_final_structure_provenance.csv`.

S12 Crystallographic descriptors and structure matching

The final CIFs were re-read independently and analysed at symmetry tolerances of 0.01, 0.05, and 0.10 Å. Table S10 gives the two endpoint assignments together with the cell dimensions. This tolerance report prevents a single symmetry setting from being treated as an exact physical classification.

Table S10: Lattice and symmetry descriptors of the four relaxed LiCr₂O₄ candidates.

Candidate	<i>a</i> (Å)	<i>b</i> (Å)	<i>c</i> (Å)	<i>V</i> (Å ³)	SG (0.01 Å)	SG (0.10 Å)
C079-1	2.946	3.824	6.706	75.547	Pm	Pmm2
C126-0	2.939	5.073	5.740	73.810	Cm	C2/m
C196-1	3.199	3.199	8.644	74.807	R3m	R3m
C234-3	3.201	3.201	8.409	74.642	P3m1	P3m1

Note: Space groups were assigned from the final CIFs with `pymatgen` at the stated symmetry tolerance. Complete lattice angles and the intermediate 0.05 Å result are retained in `SourceData/v60_crystallographic_descriptors.csv`.

Pairwise comparisons used `StructureMatcher` with $l_{\text{tol}} = 0.2$, $s_{\text{tol}} = 0.3$, an angle tolerance of 5°, primitive-cell reduction, and scale normalization. No pair matched under this recorded configuration. The six pairwise decisions are retained in `SourceData/v60_structure_matcher_pairs.csv`; this result supports structural distinctness under the stated matcher and is not a proof of different thermodynamic phases.

S13 Materials Project Mn-oxide control

The control uses a 640-candidate Mn-bearing oxide pool and Materials Project formation-energy labels. The frozen interval $[-2.59, -2.47]$ eV atom⁻¹ contains 111 targets. Corrected seeds 5–14 use

prediction-loader shuffle=False, explicit candidate-ID reindexing, batch size 16, and a 320-query budget. The element-system map contains 614 groups, including 588 singletons, and has a maximum group size of two.

Table S11 reports every corrected seed before the aggregate result is discussed. Gate and Greedy have identical batch membership and checkpoint recovery in every seed.

Table S11: Per-seed Materials Project Mn-oxide control for Gate and Greedy.

Seed	AUTC (both)	R_{80}	R_{160}	R_{240}	R_{320}	Direct	Correction
5	0.450000	26	50	81	103	19	1
6	0.450901	26	50	81	103	20	0
7	0.454505	26	50	81	106	20	0
8	0.450450	26	50	81	102	20	0
9	0.450901	26	50	81	101	20	0
10	0.450450	26	50	81	102	20	0
11	0.450000	26	50	81	99	20	0
12	0.450000	26	50	81	100	20	0
13	0.451351	26	50	81	100	20	0
14	0.450000	26	50	81	101	20	0

Note: AUTC and recovery values are identical for Gate and Greedy within each seed. “Direct” and “Correction” refer to Gate’s 20 batch decisions. The single correction in seed 5 reordered candidates within the same batch but caused no effective replacement.

The ten-seed aggregate normalized AUTC was 0.450856 ± 0.001367 for both policies. Gate retained the direct route in 199/200 batches, and no correction changed batch membership. Source-backed metrics, trajectories, pairing decisions, the full group map, and its inventory are retained in SourceData/mnoxide_corrected_seed_metrics.csv, SourceData/mnoxide_recovery_trajectories.csv, SourceData/mnoxide_seed_pairing_audit.csv, SourceData/mnoxide_element_system_groups.csv, and SourceData/mnoxide_group_inventory.csv.

S14 Development-cohort gate sensitivity

The development-cohort analysis varied the two routing thresholds on a 3×3 grid and varied each diversity weight one factor at a time around the archived setting. All configurations used the same 640-candidate pool, batch size 16, 30 MC-dropout passes, and seeds 0–4. The archived result was not retroactively changed. The best tested development value for the group-reuse penalty, $\gamma = 0.05$, was instead frozen for a separate held-out comparison; all other settings were retained. Table S12 reports every tested configuration before the aggregate interpretation in the main text.

Table S12: Complete development-cohort gate-parameter sensitivity.

Sweep	Setting	AUTC ₁₆₀	AUTC ₆₄₀	R_{80}	Correction rounds	Unique paths
Threshold	$M_0 = 0.75, G_0 = 0.40$	0.3038 ± 0.0144	0.7928 ± 0.0092	26.2 ± 2.8	39.2 ± 0.8	5/5
Threshold	$M_0 = 0.75, G_0 = 0.50$	0.3090 ± 0.0074	0.7913 ± 0.0091	27.0 ± 2.2	34.6 ± 1.3	5/5
Threshold	$M_0 = 0.75, G_0 = 0.60$	0.3038 ± 0.0117	0.7899 ± 0.0056	26.2 ± 2.3	31.0 ± 2.0	5/5
Threshold	$M_0 = 1.00, G_0 = 0.40$	0.3038 ± 0.0144	0.7928 ± 0.0092	26.2 ± 2.8	39.4 ± 0.9	5/5
Threshold	$M_0 = 1.00, G_0 = 0.50$	0.3090 ± 0.0074	0.7913 ± 0.0091	27.0 ± 2.2	34.8 ± 1.1	5/5
Threshold	$M_0 = 1.00, G_0 = 0.60$	0.3038 ± 0.0117	0.7899 ± 0.0056	26.2 ± 2.3	31.4 ± 1.3	5/5
Threshold	$M_0 = 1.25, G_0 = 0.40$	0.3038 ± 0.0144	0.7928 ± 0.0092	26.2 ± 2.8	39.4 ± 0.9	5/5
Threshold	$M_0 = 1.25, G_0 = 0.50$	0.3090 ± 0.0074	0.7913 ± 0.0091	27.0 ± 2.2	35.2 ± 1.1	5/5
Threshold	$M_0 = 1.25, G_0 = 0.60$	0.3038 ± 0.0117	0.7899 ± 0.0056	26.2 ± 2.3	32.4 ± 1.5	5/5
Weight	$\alpha = 0.05, \beta = 0.20, \gamma = 0.10$	0.3008 ± 0.0205	0.7902 ± 0.0099	25.6 ± 2.8	36.2 ± 1.3	5/5
Weight	$\alpha = 0.20, \beta = 0.20, \gamma = 0.10$	0.1977 ± 0.0463	0.7225 ± 0.0216	16.2 ± 4.8	35.6 ± 2.5	5/5
Weight	$\alpha = 0.10, \beta = 0.20, \gamma = 0.10$	0.3090 ± 0.0074	0.7913 ± 0.0091	27.0 ± 2.2	34.8 ± 1.1	5/5
Weight	$\alpha = 0.10, \beta = 0.10, \gamma = 0.10$	0.3069 ± 0.0056	0.7919 ± 0.0060	27.6 ± 1.5	35.2 ± 1.5	5/5
Weight	$\alpha = 0.10, \beta = 0.40, \gamma = 0.10$	0.3151 ± 0.0300	0.7909 ± 0.0149	27.2 ± 2.9	35.4 ± 1.5	5/5
Weight	$\alpha = 0.10, \beta = 0.20, \gamma = 0.05$	0.3292 ± 0.0087	0.7985 ± 0.0047	30.0 ± 1.2	34.8 ± 1.5	5/5
Weight	$\alpha = 0.10, \beta = 0.20, \gamma = 0.20$	0.2900 ± 0.0121	0.7822 ± 0.0078	25.0 ± 1.7	35.6 ± 1.5	5/5

Note: Values are mean $\pm$ sample SD across seeds 0–4. AUTC₁₆₀ uses the trajectory truncated at 160 queries; AUTC₆₄₀ uses the complete finite-pool trajectory. “Unique paths” gives the number of distinct complete candidate-sequence hashes among the five seeds.

Per-seed metrics, aggregate values, query-sequence hashes, the repeated-center reproducibility check, and output hashes are retained in `SourceData/v60_parameter_sensitivity_per_seed.csv`, `SourceData/v60_parameter_sensitivity_summary.csv`, `SourceData/v60_parameter_sensitivity_seed_audit.csv`, `SourceData/v60_parameter_sensitivity_reproducibility_audit.csv`, and `SourceData/v60_parameter_sensitivity_output_sha256.csv`.

S15 Same-protocol low-cost baselines

To separate the gate comparison from the historical single-run archive, three low-cost selectors were rerun on the same development cohort used for the sensitivity analysis. Energy-Gated DA-TPP, Predicted-Target Greedy, MC Dropout, and Random Sampling received the same checkpoint, matched seed policy, model schedule, batch size, query budget, and 30-pass predictive output. Their source-backed results are introduced in Table S13.

Table S13: Same-protocol development-cohort comparison with low-cost baselines.

Method	AUTC ₁₆₀	AUTC ₆₄₀	R_{80}	R_{160}	Final recovery	Unique paths
Energy-Gated DA-TPP (archived $\gamma = 0.10$)	0.3090 ± 0.0074	0.7913 ± 0.0091	27.0 ± 2.2	51.2 ± 2.5	78.0 ± 0.0	5/5
Predicted-Target Greedy	0.3159 ± 0.0355	0.7915 ± 0.0146	28.6 ± 4.2	50.6 ± 5.1	78.0 ± 0.0	5/5
MC Dropout	0.0123 ± 0.0088	0.2507 ± 0.0175	1.2 ± 1.1	2.4 ± 1.1	78.0 ± 0.0	5/5
Random Sampling	0.1131 ± 0.0221	0.4955 ± 0.0236	9.8 ± 2.3	20.2 ± 3.6	78.0 ± 0.0	5/5

Note: Values are mean $\pm$ sample SD across seeds 0–4. The Gate row uses the archived $\gamma = 0.10$ setting; the development-selected $\gamma = 0.05$ protocol is evaluated on held-out seeds in Table S14.

Candidate-level outputs and audit hashes are retained in `SourceData/v60_same_protocol_baselines_per_seed.csv`, `SourceData/v60_same_protocol_baselines_summary.csv`, `SourceData/v60_same_protocol_baselines_seed_audit.csv`,

and `SourceData/v60_same_protocol_gate_greedy_paired.csv`, together with `SourceData/v60_same_protocol_baselines_output_sha256.csv`.

S16 Frozen held-out Gate-Greedy evaluation

After the development sweep was complete, $\gamma = 0.05$ and all remaining protocol fields were frozen before seeds 15–24 were evaluated. Each seed used a different initial labelled set; both methods shared that set and the same seed-specific training and inference schedule. All 20 runs completed 640-query trajectories, and each method produced ten distinct complete sequence hashes. Table S14 reports the paired early-budget values.

Table S14: Per-seed frozen held-out comparison for Energy-Gated DA-TPP and Predicted-Target Greedy.

Seed	Gate $AUTC_{160}$	Greedy $AUTC_{160}$	$\Delta AUTC_{160}$	Gate R_{80}	Greedy R_{80}	Gate R_{160}	Greedy R_{160}
15	0.3462	0.3385	+0.0077	31	26	54	52
16	0.3269	0.2897	+0.0372	26	24	54	53
17	0.3667	0.3179	+0.0487	33	27	57	49
18	0.3103	0.2436	+0.0667	27	21	54	43
19	0.2769	0.2808	−0.0038	25	25	48	47
20	0.3064	0.3077	−0.0013	28	27	51	51
21	0.3115	0.3295	−0.0179	29	31	54	53
22	0.2974	0.3000	−0.0026	24	29	48	51
23	0.3256	0.2590	+0.0667	28	22	54	43
24	0.3551	0.3000	+0.0551	33	26	50	45

Note: $\Delta AUTC_{160}$ is Gate minus Greedy. The protocol used $\gamma = 0.05$, $M_0 = 1.00$, $G_0 = 0.50$, $\alpha = 0.10$, $\beta = 0.20$, batch size 16, and 30 MC-dropout passes.

Mean $AUTC_{160}$ was 0.3223 ± 0.0276 for Gate and 0.2967 ± 0.0297 for Greedy. The mean paired difference was 0.0256 (100,000-resample bootstrap 95% CI [0.0067, 0.0449]); the exact two-sided Wilcoxon statistic was 11 with $p = 0.105$, and paired $d_z = 0.789$. Gate led in six pairs and Greedy in four. Mean target recovery was 28.4 versus 25.8 at 80 queries and 52.4 versus 48.7 at 160 queries; both methods reached all 78 targets at 320 queries. The configuration SHA-256 begins 1e1002fcd528df18 and is retained in full with the source files. Per-seed metrics, paired differences, aggregate statistics, and output hashes are retained in `SourceData/Gamma005HoldoutAnalysis/v60_gamma005_holdout_per_seed.csv`, `SourceData/Gamma005HoldoutAnalysis/v60_gamma005_holdout_paired.csv`, `SourceData/Gamma005HoldoutAnalysis/v60_gamma005_holdout_statistics.json`, and `SourceData/Gamma005HoldoutAnalysis/v60_gamma005_holdout_output_sha256.csv`.

S17 Reproducibility inventory

The source package retains the following primary files:

- `SourceData/dft_evaluability_protocol.yaml`: frozen DFT-evaluability rules;
- `SourceData/paired_cgcnn_histories.csv`: original CGCNN acquisition histories;
- `SourceData/single_run_baseline_histories.csv`: candidate-level histories for the descriptive six-policy benchmark;
- `SourceData/Figure2_six_policy_target_recovery.csv`: plotted six-policy cumulative target counts;

- SourceData/dft_evaluability_scores.csv: DFT-evaluability probabilities and score provenance; 207
- SourceData/paired_dft_evaluability_results.csv and SourceData/paired_dft_evaluability_summary.csv: checkpoint results; 208
- SourceData/Figure3_dft_evaluability_audit.csv: plotted canonical trajectory; 209
- SourceData/v50_autc_checkpoints.csv: AUTC values at the four stopping budgets; 210
- SourceData/energy_calibration_best_oof.csv: ten outer-fold MACE-MP calibration predictions; 211
- SourceData/mlip_full_pool_results.csv: 1,280 candidate-level MLIP records and wall times; 212
- SourceData/v50_final_structure_provenance.csv and the four files in Structures/: final DFT-cell provenance and exported CIFs; 213
- SourceData/v60_crystallographic_descriptors.csv and SourceData/v60_0_structure_matcher_pairs.csv: symmetry and pairwise structure audit; 214
- SourceData/mnoxide_corrected_seed_metrics.csv, SourceData/mnoxide_recovery_trajectories.csv, and SourceData/mnoxide_seed_pairing_audit.csv: corrected Mn-oxide control; 215
- SourceData/mnoxide_element_system_groups.csv and SourceData/mnoxide_group_inventory.csv: label-blind control-pool group map; 216
- SourceData/v60_parameter_sensitivity_per_seed.csv, SourceData/v60_parameter_sensitivity_summary.csv, and the corresponding audit files: development-cohort threshold and weight sensitivity; 217
- SourceData/v60_same_protocol_baselines_per_seed.csv, SourceData/v60_same_protocol_baselines_summary.csv, and the corresponding audit files: same-protocol Gate, Greedy, MC-dropout, and random comparison; 218
- SourceData/Gamma005HoldoutAnalysis/v60_gamma005_holdout_per_seed.csv, SourceData/Gamma005HoldoutAnalysis/v60_gamma005_holdout_paired.csv, and the corresponding statistics and hash files: frozen seeds 15–24 Gate–Greedy confirmation at $\gamma = 0.05$; 219
- provenance/public_manifest_sha256.csv: repository-wide public-file hashes; 220
- SourceData/DFT_completed_candidates_report_zh.md: source report for the four completed LiCr_2O_4 candidates; 221
- Tables/generated/table_DFT_candidates.tex: exact candidate-level energy, geometry, magnetic-moment, and near-zero-gap values. 222